PSG COLLEGE OF TECHNOLOGY, COIMBATORE – 641 004

DEPARTMENT OF COMPUTER APPLICATIONS

Academic Year 2025-2026 MCA G1 & G2 Sem 4

23MX41- Project Work

Software Requirement Specification (SRS) Document

1) Title of the Project

NextDoor Connect - A Community-Oriented Digital Marketplace for Services and Resource Sharing

2) Student Detail

Roll No.	Name of the Student	Name of the Faculty Guide
24MX116	Mythily N	Ms. N. Rajeswari

3) Project Goal

The goal of the NextDoor Connect project is to develop a secure, community-based web application that enables verified residents of a residential society to offer and access services, buy or sell items, and share resources within their own community. The system ensures trust through admin verification and improves user experience with an AI chatbot assistant, promoting safe, convenient, and efficient community interactions.

4) Project Abstract

The rapid growth of on-demand service platforms has improved accessibility to home services; however, these systems often fail to address the specific needs of residential communities, such as trust, locality, and resource sharing. This project proposes NextDoor Connect, a community-based digital platform designed exclusively for neighborhoods and residential societies. The system enables verified residents to offer and request services, rent

rarely used products, and buy or sell items within their own community.

The platform incorporates admin verification by the apartment or community secretary, ensuring a high level of trust and safety. An AI chatbot assistant is integrated to understand user problems and guide them to appropriate services or products, while a smart vendor matching mechanism automatically suggests the most suitable service provider based on availability, distance, past performance, and community ratings. By promoting local skill sharing, product reuse, and secure peer-to-peer interactions, the proposed system offers a sustainable, efficient, and user-friendly solution for modern residential communities.

5) Existing System

1. Urban Company

Urban Company is a popular service aggregator platform that provides household services such as plumbing, electrical work, cleaning, and appliance repair by connecting users with professionals.

Limitations:

- Operates on a public platform without community restriction
- Service providers are not part of the residential society
- Limited trust and personalization
- No community-based interaction or resource sharing

2. OLX

OLX is an online marketplace that allows users to buy and sell second-hand products across a wide geographic area.

Limitations:

- No user or product verification

- Interaction with unknown individuals
- High risk of fraud and safety concerns
- Not designed for local or community-level transactions

3. Quikr

Quikr is a classified advertisements platform that supports buying, selling, and renting products and services.

Limitations:

- Open access to the public
- Lack of trust and controlled access
- No focus on residential societies

5. Justdial

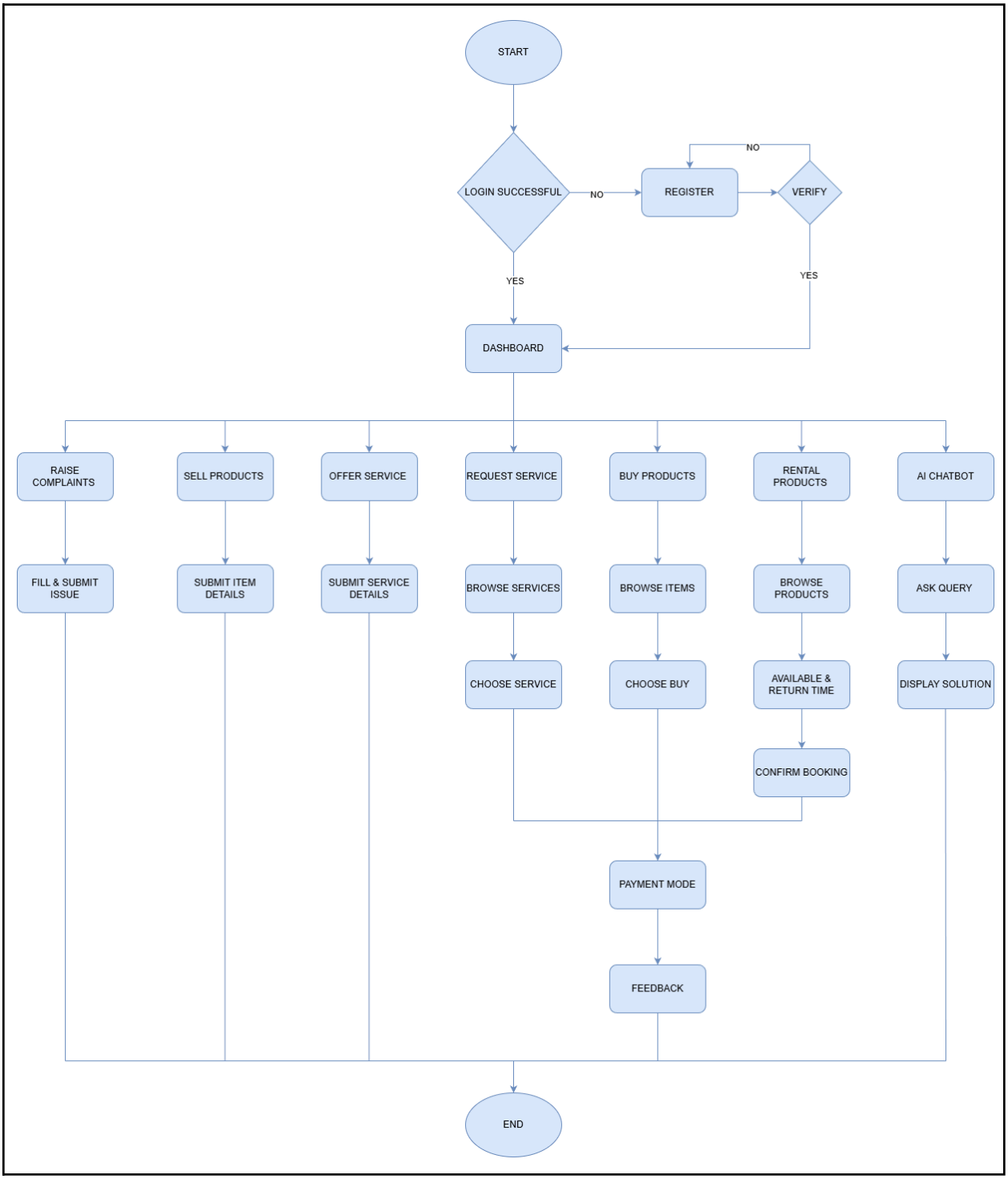
Justdial is a local search engine that provides listings of nearby businesses and service providers.

Limitations:

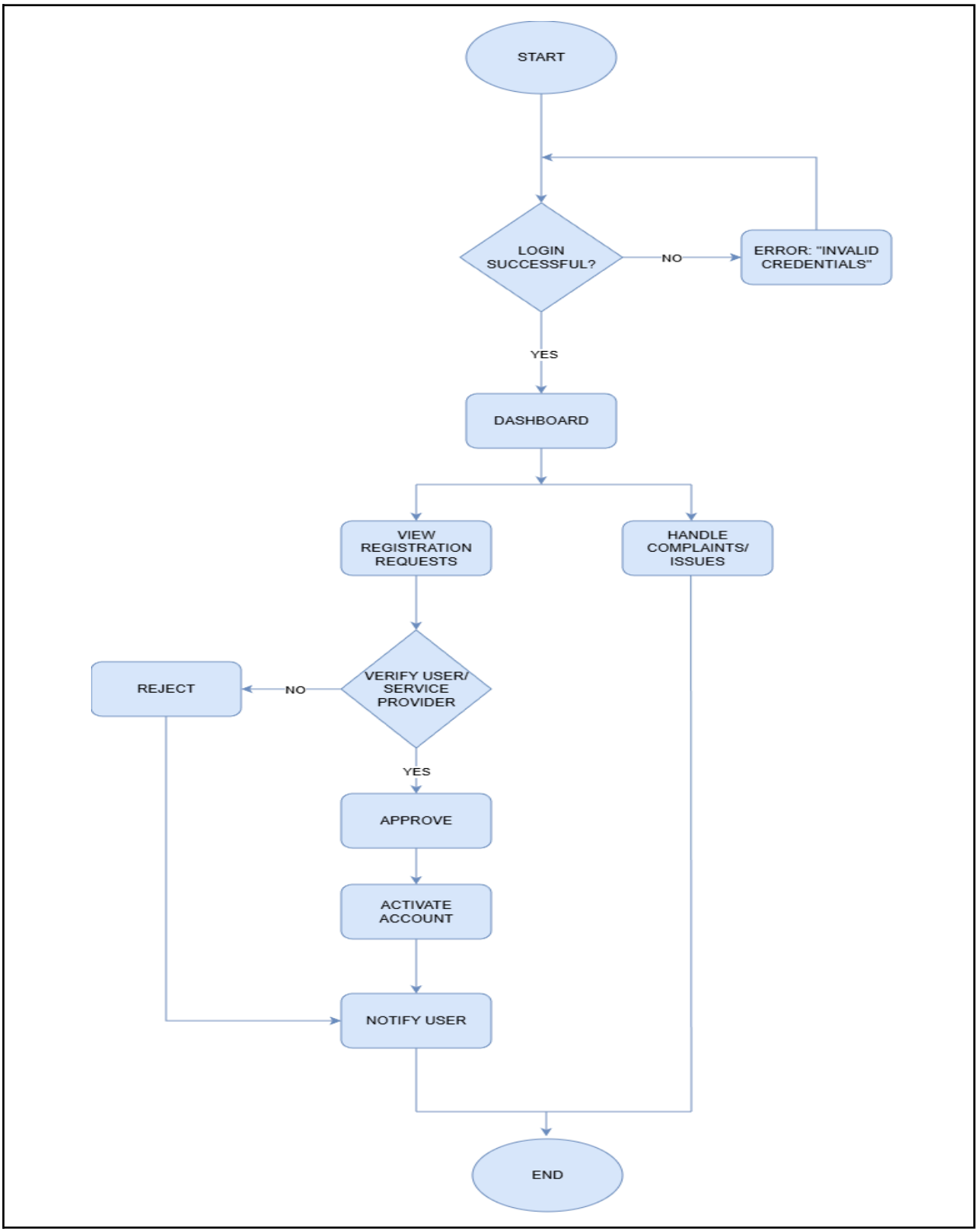
- Acts only as a directory
- No service booking or tracking
- No user authentication or verification
- Not community-oriented

6) Project Flow Analysis

User Workflow



Admin/ Community Secretary Flow



7) Outcome of the Proposed Project

- Provides a secure and trusted community-based platform for residents to access services and resources within their residential society.
- Enables verified users to buy, sell, and rent products safely, reducing fraud and security risks.
- Facilitates easy service offering and requesting through approved service providers, improving convenience and reliability.
- Strengthens community interaction and collaboration among residents by promoting local participation.
- Reduces dependency on external public platforms by offering an all-in-one solution for society-level needs.
- Ensures better transparency and accountability through feedback, ratings, and complaint management.
- Enhances user experience with an AI chatbot assistant for quick guidance and service discovery.
- Improves time efficiency and accessibility by centralizing multiple community services into a single application.

8) Modules of the Proposed Project - Functional Requirements

1. User Registration and Authentication Module

- Allows new users to register using valid details.
- Supports secure login and logout functionality.
- Provides role-based access for residents, service providers, and admin.
- Ensures only verified users can access the system.

2. Admin Management Module

- Verifies and approves registered users.

- Approves service provider requests.
- Manages user accounts and listings.
- Monitors complaints, feedback, and system activities.

3. Service Offering Module

- Allows users to apply as service providers.
- Enables approved providers to add, update, or remove services.
- Displays service availability and pricing details.
- Supports service request acceptance or rejection.

4. Service Requesting Module

- Allows users to browse available services.
- Enables users to request services based on requirements.
- Supports service confirmation and scheduling.
- Provides feedback after service completion.

5. Buy Products Module

- Allows users to browse products listed for sale.
- Enables users to view product details.
- Supports product purchase confirmation.
- Provides secure payment options and feedback.

6. Sell Products Module

- Allows users to list products for sale.
- Enables product detail and image upload.
- Allows sellers to manage product availability.

7. Rent Products Module

- Allows users to list items for rent.

- Enables users to browse rentable products.
- Supports rental duration selection and confirmation.
- Provides payment and return tracking.

8. Complaint Management Module

- Allows users to raise complaints related to services or products.
- Enables admin to track and resolve complaints.
- Provides complaint status updates to users.

9. Feedback and Rating Module

- Allows users to provide ratings and reviews.
- Helps maintain service quality and transparency.
- Displays feedback to other users.

10. AI Chatbot Assistance Module

- Provides automated responses to user queries.
- Helps in discovering relevant services or products.

9) Non-Functional Requirements

1. Performance

- The system will authenticate users and load the dashboard within 2–3 seconds under normal operating conditions.
- The application will handle multiple concurrent users within a residential society without noticeable performance degradation.
- Service listings, product listings, and search results will be displayed within 3 seconds.
- The AI chatbot will generate responses to user queries within 2 seconds.

2. Security

- The system will allow access only to admin-verified and authenticated users.
- Role-based access control will be enforced for admin, service providers, and residents.
- User credentials and sensitive data will be stored securely.
- Unauthorized access to services, products, and user data will be prevented.

3. Reliability

- The system will function continuously without unexpected failures during normal usage.
- User data such as service requests, transactions, and feedback will not be lost during system operations.
- The application will maintain consistent behavior across repeated user actions.

4. Usability

- The user interface will be simple, intuitive, and easy to navigate for all users.
- The system will require minimal learning effort for new users.
- All major functionalities will be accessible with minimal user interaction steps.

5. Availability

- The application will be available to users at all times except during scheduled maintenance.
- System downtime will be minimized to ensure uninterrupted community services.

6. Scalability

- The system will support future growth in the number of users, services, and product listings.
- New modules and features can be added without affecting existing system functionality.

7. Maintainability

- The application will be easy to maintain and update.
- Bug fixes and enhancements can be applied without disrupting system operations.

8. Compatibility

- The system will be compatible with major web browsers.
- The application will function correctly on different devices such as desktops, tablets, and mobile browsers.

10) Project Deliverables

- Software Requirement Specification (SRS) document describing system requirements and scope.
- System design artifacts including architecture diagrams, flow diagrams, and module designs.
- A fully functional NextDoor Connect web application with user, admin, and service provider modules.
- Structured database design for managing users, services, products, rentals, and transactions.
- Integrated AI chatbot for user assistance and service discovery.
- Admin panel for user verification, approvals, and system monitoring.

11) Technology to be used

- **Frontend:** React.js for building a responsive and user-friendly interface.
- **Backend:** Node.js with Express.js for handling server-side logic and API services.
- **Database:** MongoDB for storing user data, services, products, rentals, and transactions.
- **Version Control:** Git for source code management.

12) Timeline Activities

Duration	Activity
Dec 9, 2025 – Dec 20, 2025	Project topic selection, discussion with guide, finalizing project title "NextDoor Connect – A Community-Oriented Digital Marketplace for Services and Resource Sharing", preparation.
Dec 21, 2025 - Jan 9, 2026	Submission of Project Abstract, Domain study and requirement analysis. Studying existing systems such as Urban Company, OLX, Quikr, TaskRabbit, and Justdial. Finalizing functional and non-functional requirements. Drafting system flow and module design.
Jan 10, 2026	0th Review presentation conducted.
Jan 7, 2026 - Jan 20, 2026	Design of system architecture, database schema, and user flow diagrams. UI wireframe design for user, admin, and service provider modules.
Jan 21, 2026 – Feb 10, 2026	Development of frontend using React.js. Implementation of user registration, login, profile management, and dashboard modules.
Feb 11, 2026 – Feb 28, 2026	Backend development using Node.js and Express.js. Implementation of service offering, service requesting, buy, sell, and rent modules. Database integration using MongoDB.
Mar 1, 2026 – Mar 15, 2026	Integration of admin module for user verification, approvals, and complaint management. Integration of AI chatbot for user assistance.

Mar 16, 2026 – Mar 31, 2026	System testing, bug fixing, performance optimization, and validation of all modules. Preparation of project report after review from the guide.
Apr 1, 2026 – Apr 7, 2026	Final report submission, final PPT preparation, system demo setup, and viva preparation.

Signature of the Student **Signature of the Guide**